

Dirac equation 1

Consider the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = H \psi(\mathbf{x}, t)$$

The Hamiltonian H for the free-particle Dirac equation is

$$H = \boldsymbol{\alpha} \cdot \mathbf{p}c + \beta mc^2$$

where $\boldsymbol{\alpha} = (\alpha_x, \alpha_y, \alpha_z)$. The components of $\boldsymbol{\alpha}$ are the matrices

$$\alpha_x = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \quad \alpha_y = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix} \quad \alpha_z = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

and β is the matrix

$$\beta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Substitute H into the Schrodinger equation.

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = (\boldsymbol{\alpha} \cdot \mathbf{p}c + \beta mc^2) \psi(\mathbf{x}, t)$$

Rearrange as

$$\left(i\hbar \frac{\partial}{\partial t} - \boldsymbol{\alpha} \cdot \mathbf{p}c \right) \psi(\mathbf{x}, t) = \beta mc^2 \psi(\mathbf{x}, t)$$

Substitute $\mathbf{p} = -i\hbar \nabla$ for the momentum operator and move c out of the way.

$$i\hbar \left(\frac{\partial}{\partial t} + c\alpha_x \frac{\partial}{\partial x} + c\alpha_y \frac{\partial}{\partial y} + c\alpha_z \frac{\partial}{\partial z} \right) \psi(\mathbf{x}, t) = \beta mc^2 \psi(\mathbf{x}, t)$$

Left-multiply by β .

$$i\hbar \left(\beta \frac{\partial}{\partial t} + c\beta\alpha_x \frac{\partial}{\partial x} + c\beta\alpha_y \frac{\partial}{\partial y} + c\beta\alpha_z \frac{\partial}{\partial z} \right) \psi(\mathbf{x}, t) = mc^2 \psi(\mathbf{x}, t)$$

Substitute γ matrices

$$\gamma_t = \beta \quad \gamma_x = \beta\alpha_x \quad \gamma_y = \beta\alpha_y \quad \gamma_z = \beta\alpha_z$$

to obtain the free-particle Dirac equation

$$i\hbar \left(\gamma_t \frac{\partial}{\partial t} + c\gamma_x \frac{\partial}{\partial x} + c\gamma_y \frac{\partial}{\partial y} + c\gamma_z \frac{\partial}{\partial z} \right) \psi(\mathbf{x}, t) = mc^2 \psi(\mathbf{x}, t)$$

In spacetime notation with $c = 1$

$$i\hbar \left(\gamma^0 \frac{\partial}{\partial x^0} + \gamma^1 \frac{\partial}{\partial x^1} + \gamma^2 \frac{\partial}{\partial x^2} + \gamma^3 \frac{\partial}{\partial x^3} \right) \psi(x^\mu) = m\psi(x^\mu)$$

Verify dimensions.

$$\hbar \frac{\partial}{\partial t} = [\text{J s}] [\text{s}^{-1}] = [\text{J}]$$

$$\hbar c \frac{\partial}{\partial x} = [\text{J s}] [\text{m s}^{-1}] [\text{m}^{-1}] = [\text{J}]$$

$$mc^2 = [\text{J}]$$

Eigenmath script